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Scenario Modeling for Urban Road Emergency Management: A Cognitive Digital Twin–Based Method

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Abstract: The cognitive digital twin (CDT) concept provides a promising way to organize and integrate the full life-cycle data of urban road emergency scenarios, as well as facilitate scenario deduction and prediction. The construction of twin models constitutes the prerequisite for the practical implementation and application of CDT. However, the existing research on emergency scenario twin modeling falls short in providing a unified modeling method that comprehensively captures the multidimensional characteristics of scenario elements. This deficiency, coupled with inadequate analysis of interelement interactions and constraints posed by semantic inconsistencies and interoperability among twin models, hampers the accurate mapping of physical scenarios and their evolution dynamics. In response to the multielement, uncertainty, and dynamic nature of urban road emergency scenarios, this paper proposes a CDT modeling method for urban road emergency scenarios. First, the definitions of element-level and scenario-level twin models are elucidated in three dimensions—geometry, attribute, and behavior—in conjunction with resilience theory. Second, a modeling method based on SysML and ontology is proposed for twin models at different levels, and a SysML2Ontology transformation tool is developed to enable the systematic construction of emergency scenario cognitive digital twin (ESCDT). Furthermore, a CDT framework for urban road emergency management centered on twin models is introduced. Finally, the North-South Elevated Road in Shanghai, China, is used as the engineering application object. The feasibility of our modeling methodology is validated through a case study, which involves the construction of the ESCDT for a collision accident. The model can be further utilized for emergency scenario deduction, ultimately elevating the effectiveness of emergency management. **DOI: 10.1061/JCCEE5.CPENG-6190.** © *2025 American Society of Civil Engineers.*

Introduction

With the rapid growth of the world's population and the acceleration of urbanization, vehicles and traffic flows are soaring, increasing traffic emergencies such as bad weather, traffic accidents, or road failures. They can lead to disruptions in the operation of the transportation system, placing great pressure on the reliable travel and robust operation of urban roads (Stefaniec et al. 2020), directly threatening the resilience of urban road systems (Osman 2021). The resilience of urban road systems is generally considered to be the ability to absorb, adapt, and quickly recover after being disturbed (Bešinović 2020). When a traffic accident occurs, resilience is mainly ensured through technical and organizational means, that is, urban road emergency management (Yu et al. 2024). Effective urban road emergency management for enhanced resilience necessitates the integration of the unique characteristics of the urban road during traffic accidents, real-time situational awareness, and foresight of potential accidents. (Ford and Wolf 2020). To satisfy these requirements, scholars have been applying digital twin (DT) technology to improve the resilience of urban transportation systems (Miglani and Kumar 2019). Braik and Koliou (2023) proposed a DT framework for emergency management of transportation infrastructure during natural disasters that focuses on data acquisition, analysis, and reasoning, integrating physics-based and data-driven models to simulate and predict facility performance and assisting in the adjustment of emergency strategies and planning. Feng et al. (2023) analyzed the data sharing problem in the Internet of vehicles transportation system from a DT perspective, developed travel behavior models under uncertain events, and established a blockchain-enabled DT transportation system to improve traffic network resilience. However, the urban road system is complex, consisting of people, vehicles, roads, and environments. The interactions between various elements during traffic accidents are uncertain and dynamic, which challenge the ability to effectively model, understand, and predict system performance (Alibrandi 2022).

Cognitive digital twin (CDT) empowers DT with enhanced cognitive capabilities and realizes DT integration with the help of semantic technologies such as ontology and knowledge graph (Zheng et al. 2022), which facilitates the integration of life-cycle data, information, and knowledge of urban road emergency scenarios. CDT has been studied in intelligent manufacturing (Eirinakis et al. 2022), smart cities (Liu et al. 2023), etc. Similar to DT, CDT takes the twin models as the core, and the division and definition of the models have distinct domain characteristics.

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When urban road emergencies occur, twin modeling of emergency scenarios requires recognizing the complex dynamic coupling relationships between multiple scenario elements and focusing on the chain process of scenario evolution and its mutation characteristics. In response to addressing the issue of completeness in twin modeling, this paper proposes the multilevel and multidimension definition for emergency scenario twin models, informed by resilience theory and accounting for the multifaceted, uncertain, and dynamic nature of urban road emergency scenarios. Subsequently, in view of the characteristics of twin models at different levels and dimensions, a systematic method of model construction and integration is proposed, aimed at unraveling the underlying mechanisms governing the interactions among elements. Finally, with the twin model as the core, a CDT framework is proposed for urban road emergency management.

The remaining part of this paper is organized as follows. First, the research related to DT-based emergency management is reviewed in terms of both models and modeling approaches, and the research gap is identified. The next section introduces and meticulously discusses the CDT model construction and integration method for urban road emergency scenarios, together with an exposition of the CDT framework tailored for this setting. Then, an emergency scenario twin modeling case under a collision accident is introduced to verify the proposed method. Finally, the results and future directions of the study are discussed and summarized.

Related Work

DT provides a new way to synchronize the physical and virtual worlds (Grieves 2014). In the field of emergency management, DT leverages high-fidelity models and dynamic simulations updated in real time to help decision-makers generate insights into current scenario during and after the emergency occurrence, thus enabling enhanced emergency management and significantly increased resilience (Braik and Koliou 2024b). However, existing research shows that ensuring consistency and interoperability of models is both important and challenging for DT-based emergency management (Deng et al. 2021). To address this challenge, recent studies have combined semantic modeling and system engineering with DTs to improve interoperability and connectivity among twin models (Rožanec et al. 2022). On this basis, the concept of CDT has been introduced (Adl 2016), and recent works have explored the potential of CDT in smart manufacturing systems. Nevertheless, in the field of emergency management, there remains a need for further investigation into the CDT model and associated modeling methodologies, which are the core of the CDT application.

Twin Models for Scenario-Based Emergency Management

DT can be defined as a multiphysics, multiscale model where digital models are simulated, analyzed, and updated based on data obtained from the physical world (Longman et al. 2023). Therefore, the foundation of DT-based emergency management is the twin modeling of emergency scenarios in the physical world. Classical disaster theory identifies hazard elements, affected elements, and environment elements as the constituent elements of emergency scenarios in the physical world (Blaikie et al. 2014). The hazard element and affected element, respectively, denote the elements causing the emergency and those affected by it, while the environment element serves as the backdrop conditions that affect them. Moreover, response plans outline action schemes aimed at mitigating the impact of an emergency (Ni et al. 2023), their implementation driving system recovery from the emergency's effects, and these plans themselves becoming pivotal elements driving the evolution of scenarios (Wang et al. 2020). Thus, the hazard element, affected element, environment element, and response plan element constitute the key physical objects in constructing an emergency scenario twin model. The interrelationships among these various scenario elements drive the evolution of emergency scenarios. Therefore, the construction of a twin model based on the interaction mechanism between the elements of the emergency scenario can provide a panoramic description of the emergency scenario (Kang et al. 2023).

The core of DT is a high-fidelity model that contains a multidimensional fusion of geometry models, attribute models, behavior models, and rule models, so it can truly depict and simulate the state of physical entities (Tao et al. 2019). In the field of emergency management, Li et al. (2018) analyzed the dynamic interactions between flood inundation and traffic flow and conducted real-time simulation modeling of flood scenarios from the geometric and behavior dimensions to predict traffic disruptions. Lv et al. (2022) used traffic accident intelligent algorithms to analyze the attribute characteristics of involved elements, establishing a traffic accident prevention and prediction system driven by DT and artificial intelligence. To enhance the understanding of the full-stage and overall posture of emergency scenarios (Arvin et al. 2023), resilience-based scenario analysis and modeling methods have gradually gained attention. Resilience embodies a system's and its subsystems' capacities to absorb, adapt to, and recover from adverse or abnormal states, serving as an effective means of integrating information about scenario elements and fostering a holistic comprehension of emergency scenarios (Field et al. 2022). Jiao et al. (2021) established a storm disaster scenario model by identifying the key factors influencing the resilience of urban rail transit stations and their action paths. Yu et al. (2023) applied the pressure-state-response theory to analyze the interaction mechanisms between multidimensional scenario elements and the characteristics of the resilience stage and established an emergency scenario model from a resilience perspective. Resilience provides a new perspective for in-depth exploration of the coupling relationship and dynamic evolution pattern among scenario elements. Therefore, this paper considers that it is promising to integrate multidimensional scenario element information within the resilience framework to construct multilevel twin models of emergency scenario cognitive digital twin (ESCDT).

Twin Modeling Approach for Emergency Scenario

Appropriate modeling methods facilitate the practice of twin models. Multiple studies have been carried out to explore the construction of emergency scenario twin models. For the construction of geometry models, White et al. (2021) integrated building information modeling (BIM), big data, and IoT sensor data to create a DT-supported smart city, in which BIM was used to present the collected information in 3D format. For the construction of attribute models, Liu et al. (2020) collected indoor real-time operating information to extract the key attributes of the building and used machine learning models to automate the classification and severity assessment of emergency scenarios. For the construction of behavior models, Fang et al. (2023) introduced a structural model to simulate the safety performance of workers' behavior under different intervention strategies by considering the behavioral decision-making process under emergency scenarios and the interaction between workers and the environment.

Existing research predominantly proposed modeling methods for a single dimension in a certain application field. However, an emergency scenario contains multiple elements, spans multiple phases, and is a complex object with uncertainty, which requires a systematic modeling approach to construct and integrate models in different dimensions. SysML is a widely used system modeling

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language to graphically describe different aspects of the system such as structure, attributes, and behaviors (Ramos et al. 2012). Wang et al. (2021) utilized *block definition diagrams* to integrate DT with SysML for analyzing and predicting system design complexity. Wang et al. (2023) constructed a DT model for autonomous vehicles using *block definition diagrams* and *state machine diagrams*, along with *requirement diagrams* to identify driving scenarios. SysML v2 is developed to address the limitations of SysML v1 (Bajaj et al. 2022). Its *definition package* contains packages to define different kinds of elements, such as the *part definition*, *item definition*, *action definition*, and *attribute definition*. Therefore, SysML models can effectively capture the constituent parts and multifaceted characteristics of systems. However, the SysML models have a limited semantic basis and are not computer-readable, resulting in poor interoperability between models (Arista et al. 2023).

Ontology is often used to solve semantic issues because it enables modeling data in a unified form and introduces the possibility of data expansion (Chen et al. 2020). Specifically, it represents knowledge as a set of concepts within a domain, using a shared vocabulary to represent the types, attributes, and interrelationships of these concepts, providing an effective way to integrate heterogeneous data, information, and knowledge involved in the CDT models (Zheng et al. 2022). Jia et al. (2022) proposed a generic ontology-based information fusion method that can construct complex DT models with multiscale, multidimensional, and multiscenario information. Jiang et al. (2023) developed a building firefighting ontology to fuse static geometry information and dynamic sensory data and deployed this ontology as the DT model to help respond quickly to fire events.

Relevant studies indicate that SysML and ontology are the main enabling technologies for CDT modeling and integration. In this study, we consider using SysML and ontology for twin modeling of urban road emergency scenarios to solve the problem of representing and integrating multilevel, multidimensional information.

Knowledge Gaps

The review of existing literature highlights the following knowledge gaps:

1. The existing research on emergency scenario twin models predominantly focuses on the geometrical modeling of scenario elements, overlooking the definition and construction of other dimensions such as attribute and behavior models. The absence of attribute models hinders the description of crucial characteristics of scenario elements, thereby impeding an accurate reflection of scenario state. Similarly, the lack of behavior models precludes the depiction of the evolutionary behaviors and interactions of scenario elements, making it challenging to illustrate the dynamic evolution of the scenarios. Consequently, the failure to precisely construct and integrate multidimensional models of scenario elements results in an inadequate mirroring of their real-world states, hindering the precise mapping of emergency scenarios.
2. The urban road emergency scenario is a complex object, and while geometric modeling methods for scenario elements have matured, the construction of attribute models has not yet fully established the mapping between the attribute models of scenario elements and scenarios. The construction of behavior models has also not delved into behavioral traits, response mechanisms, and evolution patterns. Moreover, the analysis and construction of the interactions and coupling relationships among scenario elements are crucial for the assembly and integration of element-level models. However, there is a lack of research on the assembly and integration of models from a multilevel perspective, ranging from element-level to scenario-level. Additionally, models exhibit deficiencies in semantic consistency and interoperability, limiting their effective CDT-based applications such as emergency simulation and decision-making.

Therefore, this study proposes a method to define, construct, and integrate CDT models of urban road emergency scenarios. First, this article combines resilience theory to define multidimensional and multilevel twin models to achieve a complete depiction of emergency scenarios. Further, based on the characteristics of twin models at different levels and dimensions, a model construction and integration method based on SysML and ontology is proposed, and a model transformation tool is developed, thereby facilitating the systematic and automatic construction of ESCDT. Building upon the constructed twin models, the study introduced a CDT framework for urban road emergency management.

Methodology

Models for ESCDT

In this paper, the twin models for urban road emergency scenarios are classified into element-level and scenario-level, and different-level twin models (LTMs) are constructed from multiple dimensions such as geometry, attribute, and behavior, as shown in Fig. 1.

Among them, the geometry model depicts the external manifestation of emergency scenarios with information such as the geometric appearance and geometric structure of elements. The attribute model (AM) is used to describe the characteristics of elements in the scenario evolution process, including feature attribute (FA) and association attribute (AA). FA refers to the inherent attributes of an element, whereas AA denotes attributes that arise from physical linkages or connections with other elements (called associated elements). For example, vehicle cargo and vehicle hazard element have associated attributes such as *loadedOnVehicle* and *loadedCargo*, respectively, which shows a strong linkage between them. Therefore, AA portrays the physical interdependencies between elements (Rinaldi 2004). Such interdependencies can be classified into compositional and noncompositional interdependencies. Compositional interdependencies denote that the associated element is a component of the current element, and then the corresponding AA defines the information such as function and structure related to the associated element. The AA under noncompositional interdependencies describes the interactions of the elements, the degree of influence, and the way they are associated. Changes in the attributes of elements are driven by certain behaviors, so the behavior model (BM) is employed to elucidate the interactions between elements. BM involves four components, including the behavior element (BE), behavior condition (BC), behavior logic (BL), and behavior result (BR). BE is the set of scenario elements involved in behavior. BC is the condition under which the behavior occurs, consisting of a series of propositions that determine the attributes of the involved elements. BC denotes receiving input from other elements, so BM can characterize the functional interdependencies between elements (Aghababaei and Koliou 2022). BL is the logic of behavior and describes the process of attribute changes. BR embodies the outcome of attribute alterations resulting from the behavior. Employing the BM enables the portrayal of individual element evolution and interdependencies, thereby elucidating the logical relationships among elements within emergency scenarios.

Building upon multidimensional models encompassing geometry, attribute, and behavior models, the element-level twin model

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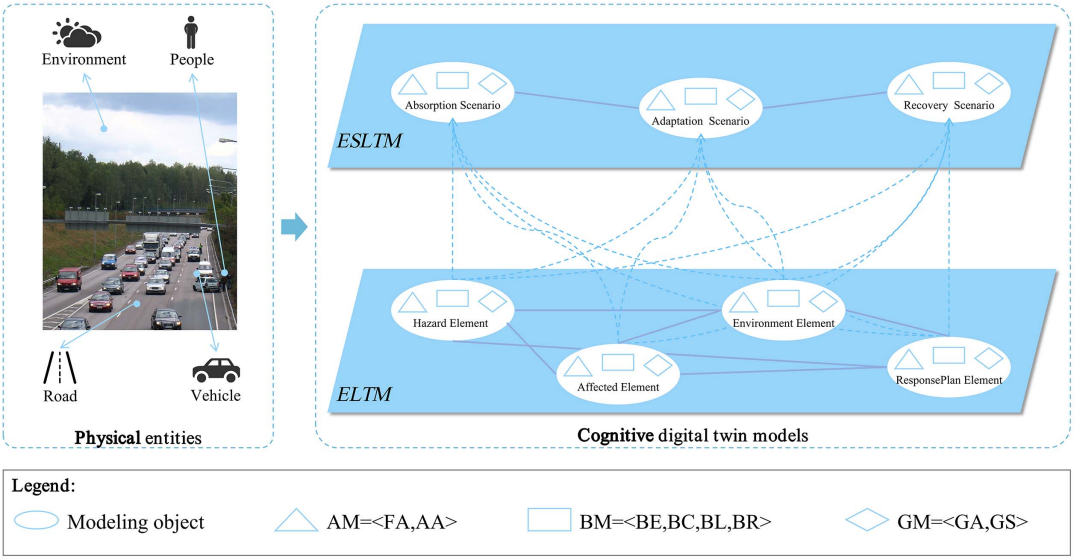


Fig. 1. CDT models for ESCDT.

Table 1. Specific elements in urban road emergency scenario

Scenario element classification	Specific element	Description
Hazard element	Vehicle hazard element	The vehicle that caused the accident
Affected element	Vehicle affected element	The vehicle that was subjected to the accident
	People affected element	The injured passenger that was affected by the accident
	Road affected element	The road that was affected by the accident
Environment element	Road environment element	The road where the accident occurred
	Meteorology environment element	The meteorological conditions under which the accident occurred
Response plan element	Response action element	Actions such as fire extinguishing, rescuing the injured, and emergency rescue to reduce the impact of accidents and restore the functions of the transportation system
	Response resource element	The resources used in the implementation of emergency actions include personnel, vehicles, and materials, such as ambulance personnel and ambulance vehicles, firefighters, and fire-fighting vehicles.

(ELTM) establishes digital representations of hazard elements, affected elements, environment elements, and response plan elements within physical emergency scenarios, thereby creating virtual mappings of each element and its interrelationships. Table 1 presents specific elements within urban road emergency scenarios. Furthermore, the intricate interaction and dynamic evolution of multiple elements render precise comprehension and predictive analysis of such scenarios highly challenging. Therefore, integrating different ELTMs to construct an emergency scenario-level twin model (ESLTM) is crucial for the full-cycle emergency management of urban roads. Existing scenario analysis methods often depict coupling relationships between elements from a single stage and a single dimension; to improve the dynamics and integrity of emergency scenario cognition, this paper comprehensively considers the multidimensional characteristics and interrelationships of scenario elements to create ESLTM based on resilience theory, providing a digital representation of emergency scenarios. Referring to the resilience framework proposed by Yarveisy et al. (2020), emergency scenarios are divided into absorption scenarios, adaptation scenarios, and recovery scenarios, which correspond, respectively, to the pre-event, during-event, and postevent phases of an emergency's life cycle. To capture the multidimensional characteristics of emergency scenarios, the AMs of ESLTM are first defined by extracting safety, function, time, and economy attributes from the AMs of ELTM. Subsequently, the BMs of the ESLTM are formed by integrating the BMs of ELTM into a panoramic view of scenario evolution, guided by the concurrent and sequential relationships among individual behaviors.

Modeling for ESCDT

A systematic modeling method with high fidelity, punctuality, and efficiency is an essential support for the CDT model. Existing relevant research shows that BIM is the most commonly used geometric modeling technology, and its application has been extensively explored (Pan and Zhang 2021). On account of this, a novel CDT modeling method is proposed based on SysML and semantic technology, constructing and integrating AMs and BMs to build different LTMs.

Modeling of ELTM

The modeling of ELTM is the fundamental component of the ESCDT, which is responsible for providing methods for the description of the characteristics of different elements. The modeling focus of attributes is diverse for various types of elements. For hazard and affected elements, type, position, condition parameters, and other describable attributes are considered significant information

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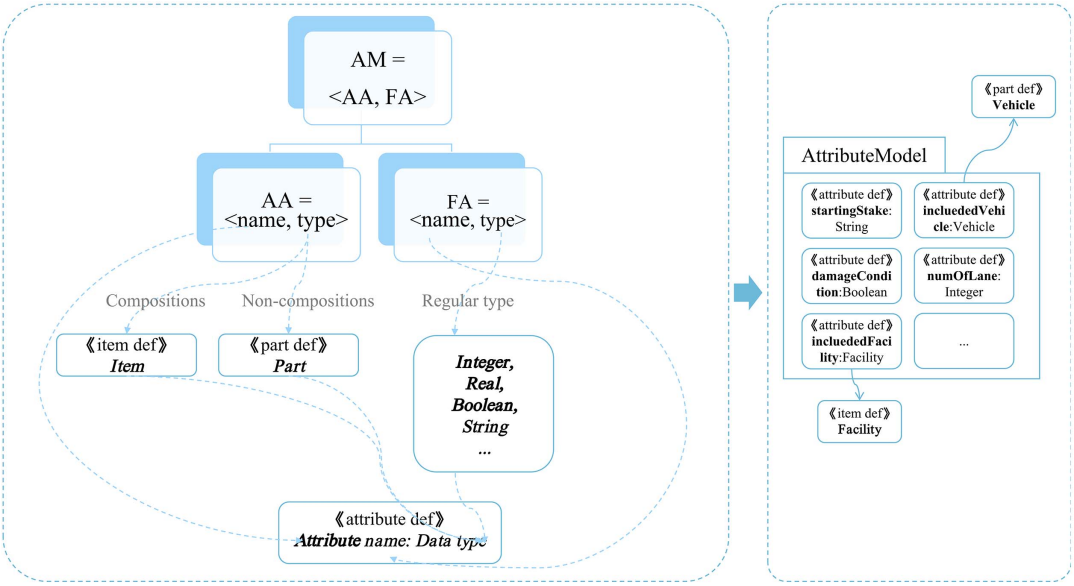


Fig. 2. Construction of AM based on *attribute definition*.

for tracking the evolution of the emergency scenario. Similarly, environment elements, characterized by meteorological parameters, affected areas, and the condition, play a pivotal role in driving scenario evolution. In the context of response plan elements, attributes such as the types, utilization, and execution status of emergency resources and actions significantly influence the emergency scenario. This study employs SysML v2's *attribute definition* to construct AMs for the elements, as illustrated in Fig. 2. The attribute defines the data type of a set of data values, which can be expressed as

$$\text{Attribute} = <\text{attribute name}, \text{data type}>$$

The attribute name is denoted by *attribute name*, while the data type of an attribute is determined based on its specific attribute and represented by *data type*. For AA, the data type selection is guided by the type of association between the element and the associated element. If the association is compositional, an *item definition* is referenced to denote the associated element; otherwise, a *part definition* is used to represent it. The data type of FA can directly leverage standard data types provided by SysML v2, such as *integer*, *real*, *boolean*, or *string*.

In the modeling of behavior, *action definition* and *state definition* utilize respective parameters and states to portray actions, as illustrated in Fig. 3. Specifically, BC and BR serve as the *in parameter* and *out parameter* in *action definition*. The evolving attributes of elements mirror their respective states. Hence, elements and their attributes from the BE and BL are transformed into states, which are then represented by state nodes and transitions within a state machine diagram within the *state definition*.

ELTM can be constructed by integrating the above multidimensional models. It can be represented by *part definition*, which mainly includes attributes, perform actions, exhibit states, and items, as shown in Fig. 4. These three constituents are respectively a usage of the previously defined *attribute definition*, *action definition*, and *state definition* and are used here as certain aspects of the *part definition*.

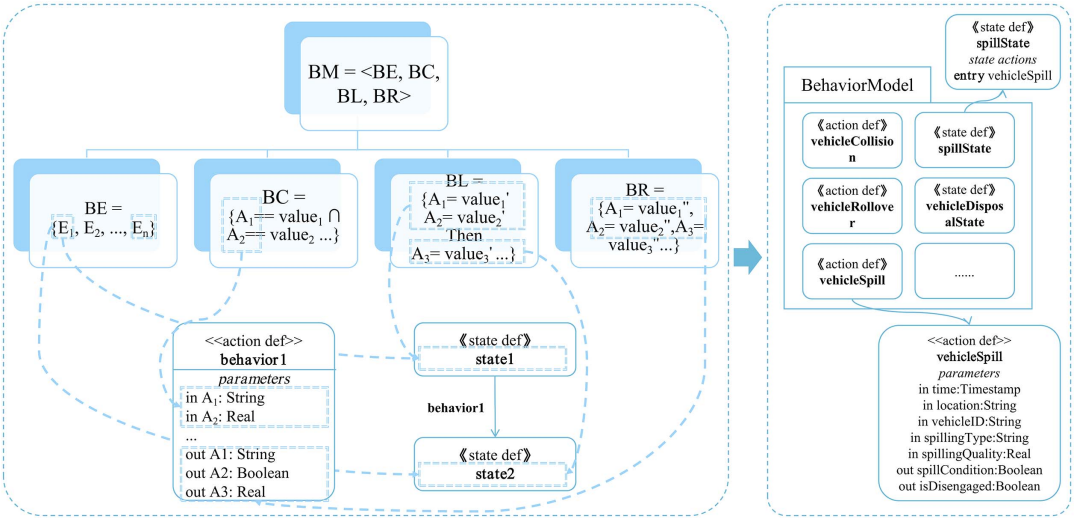


Fig. 3. Construction of BM based on *action definition* and *state definition*.

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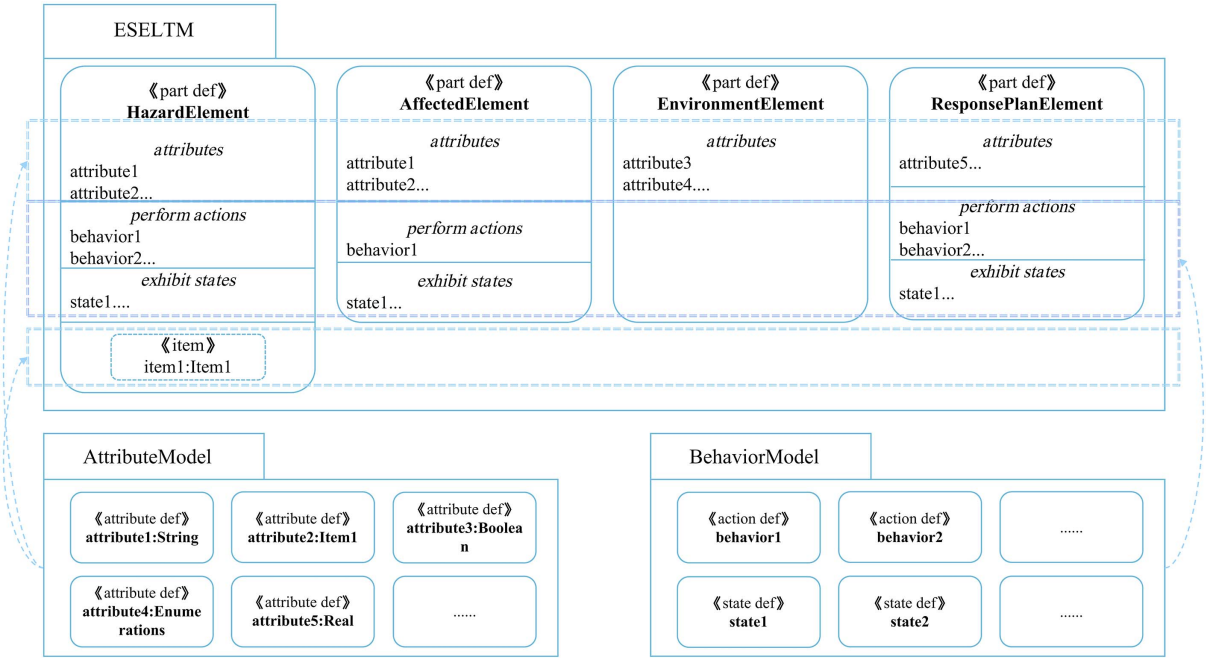


Fig. 4. Construction of ELTM based on SysML.

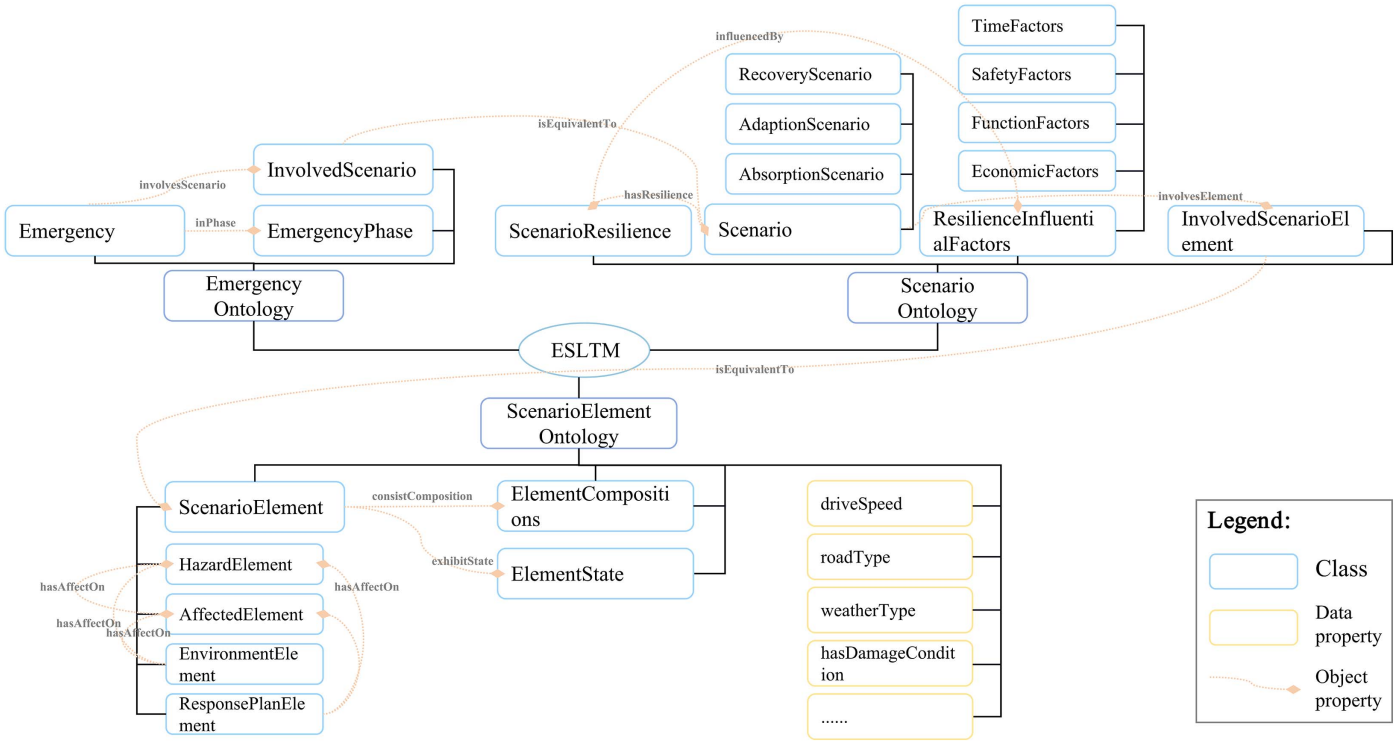


Fig. 5. Ontology base of ESLTM.

Modeling of ESLTM

As mentioned in the section "Models for ESCDT," ESLTM integrates ELTMs based on their interactions and coupling relationships. However, SysML models are inherently limited in terms of semantic consistency and interoperability, which may lead to information ambiguity and increased complexity when directly combined. In contrast, ontology models, with their rigorous logical structures and robust reasoning capabilities, excel at clearly organizing and expressing the coupling relationships among ELTMs, as well as facilitating the understanding and analysis of underlying scenario characteristics. Consequently, this section presents an ontology-based ESLTM construction method; the ontology base contains three core ontologies: the *Emergency* ontology, the *Scenario* ontology, and the *ScenarioElement* ontology, as shown in Fig. 5.

1. The *Emergency* ontology represents an emergency, which can be viewed as a collection of emergency scenarios. The *Emergency* class, as the core class, is associated with other classes (e.g., *InvolvedScenario*, *EmergencyPhase*) that

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characterize it through object properties. For example, *inPhase* indicates the relationship between *Emergency* and *EmergencyPhase*.

2. The *Scenario* ontology details the structure and situational characteristics of emergency scenarios. It includes the *Scenario* class and its subclasses: *AbsorptionScenario*, *AdaptionScenario*, and *RecoveryScenario*. Additionally, the *ScenarioResilience* describes the resilience characteristic of emergency scenarios, which are influenced by multiple factors. These factors are represented by *ResilienceInfluentialFactors*, which contains four subclasses: *SafetyFactors*, *FunctionFactors*, *TimeFactors*, and *EconomicFactors*. Similarly, these classes are associated with the *Scenario* class through object properties (e.g., *involesElement* between *Scenario* and *ScenarioElement*).
3. The *ScenarioElement* ontology is the ontological expression of the ELTM in the section "Modeling of ELTM," serving as the foundation for the above two ontologies. The *ScenarioElement* class contains four subclasses: *HazardElement*, *AffectedElement*, *EnvironmentElement*, and *ResponsePlanElement*. The element components involved in associative attributes are represented by *ElementCompositions*, so associative attributes can be represented as the *consistsComposition* object property. The feature attributes are represented by the corresponding data properties (e.g., *driveSpeed*, *hasDamageCondition*, *WeatherType*). The behavior model is transformed into object properties. For example, *hasCollisionAffectOn* indicates the collision between *HazardElement* and *AffectedElement*.

The *Emergency* ontology, the *Scenario* ontology, and the *ScenarioElement* ontology are correlated by object properties. The *InvolvedScenario* class of the *Emergency* ontology is equivalent to the *Scenario* class of the *Scenario* ontology. The *InvolvedScenarioElement* class of the *Scenario* ontology is equivalent to the *ScenarioElement* class of the *ScenarioElement* ontology. Finally, these three correlated ontologies form ESLTM.

Modeling of ESCDT

According to the definition in the section "Modeling of ESLTM," ESLTM comprises three constituent ontologies: the *Emergency* ontology, the *Scenario* ontology, and the *ScenarioElement* ontology. To realize the systematic construction of ESCDT, a novel approach is proposed in this section, which first proposes the transformation rules and algorithms for converting SysML v2-based ELTM to the *ScenarioElement* ontology and extracts the relevant instance information. Second, corresponding ontologies are generated based on the definitions of the *Scenario* ontology and the *Emergency* ontology; the corresponding individuals of the two ontologies are obtained by reasoning semantic web rule language (SWRL) rules.

Based on the basic element categories of the ontology model, the transformation rules (see Fig. 6) can be divided into three major categories: class, property, and relationship. The owl classes are developed based on the interested entities, so they can be obtained through the conversion of *part definition* and *item definition*, which represent such entities, as shown by Rules 1 and 2. Furthermore, *state definition* is also translated into the owl classes, as illustrated by Rule 3. The data properties are developed to describe the attributes of the entities, and, thus, they can be generated from *attribute definition*, which encapsulates the entity's FA, as demonstrated by Rule 4. The object properties are developed to describe the interrelationships between entities. These are derived from *action definition* and *state definition*, which describe the entity's BM, as well as *item definition* and *part definition*, which capture the entity's AA, as shown in Rules 5–9.

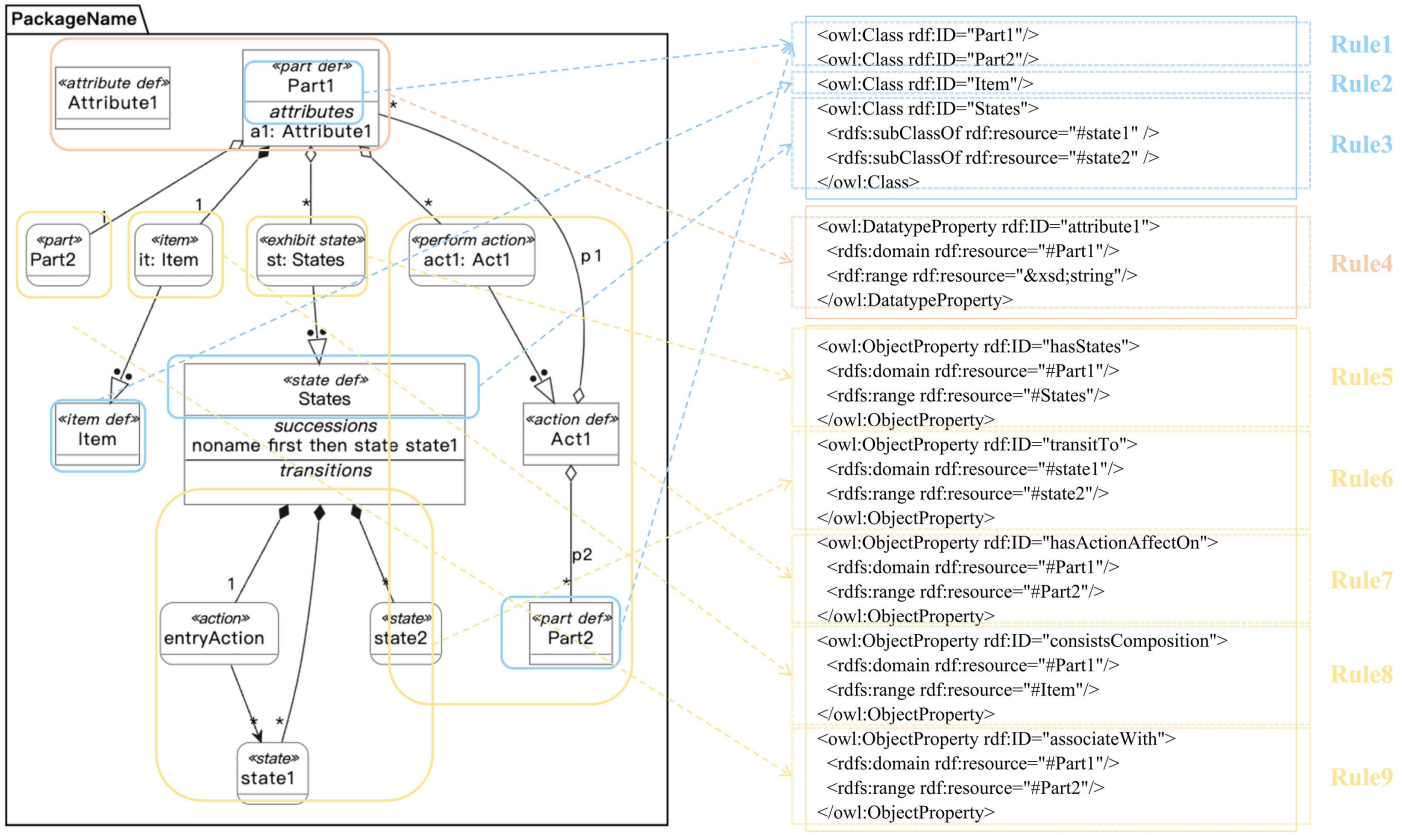


Fig. 6. Model transformation rules.

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Based on the above transformation rules, the corresponding model transformation method is summarized in the following algorithms.

According to the first three rules, Algorithm 1 (see Fig. 7) converts *part definition*, *item definition*, and *state definition* utilized in ELTM to the corresponding classes in the ontology depending on their type. Once the owl classes are obtained, the individuals of the corresponding classes are generated.

In Algorithm 2 (see Fig. 8), part of the *attribute definition* referenced by the *part definition* is converted to data properties in the ontology by traversing them following Rule 4. Additionally, the corresponding property value of the individuals can be obtained.

According to Rules 5–9, Algorithm 3 (see Fig. 9) first extracts the objects associated with the *action definition*, the *state definition*, and the rest of the *attribute definition*, and then they are transformed into object properties.

Building upon the individuals of the *ScenarioElement* ontology derived from the above transformation, according to the *Scenario* ontology and *Emergency* ontology structures defined in the section "Modeling of ESLTM," and their relationship with *ScenarioElement*
ontology, the real-time individuals of the two can be dynamically generated through SWRL rule-based reasoning, as outlined in Table 2.

A SWRL rule comprises an antecedent and a consequent, both of which contain positive conjunctions of atoms. The atoms can be of the form *Class(?x)*, *Class(?y)*, *Attribute(?x, ?a)*, or *Relation(?x, ?y)*, where *Class* is a class, *Attribute* is a data property, *Relation* is an object property, and *?x* and *?y* are variables, individuals, or data values. In this paper, the rules can be divided into association rules between scenario element and scenario, and association rules between scenario and emergency.

Algorithm 1 The algorithm for class transformation

Input: *Part Definition*, *Item Definition*, and *State Definition* of ELTM
Output: Classes of the *ScenarioElement* ontology
*/*Apply Class Transformation Rule*/*
*/*ClassTransformation(P)*/*

```
for each Part P in SysML Document do
    if Class P not exist then
        Apply Rule 1:PartToClass Transformation Rule on P
        Declare an individual of P
        for each feature in P.features do
            if feature.type == "items" then
                for each Item I in items do
                    if Item Class not exist then
                        Apply Rule 2:ItemToClass Transformation Rule on I
                        Declare an individual of I
                        PropertyTransformation(I)
                    end if
                end for
            end if
        end for
        PropertyTransformation(P)
    end if
end for
for each State St in SysML Document do
    if Class st not exist then
        Apply Rule 3:StateToClass Transformation Rule on St
        Declare an individual of St
    end if
end for
```

Fig. 7. Algorithm 1 for class transformation.

Algorithm 2 The algorithm for property transformation

Input: *Part Definition* of ELTM
Output: Data properties of the *ScenarioElement* ontology
*/*Apply Property Transformation Rule*/*
*/*PropertyTransformation(Entity)*/*

```
for each Entity in Part.Definition do
    if Entity.type == "attributes" then
        for each feature in Entity.features do
            if feature.type == "attributes" then
                for each attribute Att in feature do
                    if Att.DataType in RegularDataTypeList then
                        Apply Rule 4:RegularAttToDataProperty Transformation Rule on Att
                        Declare property value to the individual of Entity
                    end if
                end for
            end if
        end for
    end if
    if feature.type == "perform actions" then
        for each action Act in feature do
            RelationshipTransformation(Act, "Action")
        end for
    end if
    if feature.type == "exhibit states" then
        for each state State in feature do
            for each state State in feature do
                RelationshipTransformation(State, "State")
            end for
        end for
    else
        RelationshipTransformation(features, "AssociateAttribute")
    end if
end for
```

Fig. 8. Algorithm 2 for property transformation.

Algorithm 3 The algorithm for relationship transformation

Input: *Action Definition*, *State Definition*, and *Attribute Definition* of ELTM
Output: Object properties of the *ScenarioElement* ontology
*/*Apply Relationship Transformation Rule*/*
*/*RelationshipTransformation(rel, type)*//*

```
for each rel in RelationshipDefinition do
    if type == "State" then
        stateOwner = rel.ParentPart
        stateTransition = GetTransition(rel)
        Apply Rule 5:ExhibitStateToObjectProperty Transformation Rule on stateOwner And rel
        Declare relationship between individual of stateOwner and individual of rel
        Apply Rule 6:StateTransitionToObjectProperty Transformation Rule on stateTransition.sourceState And stateTransition.targetState
        Declare relationship between individual of stateTransition.sourceState and individual of stateTransition.targetState
    else if type == "Action" then
        relStart = GetInParamOf(rel).ParentPart
        relTarget = GetOutParamOf(rel).ParentPart
        Apply Rule 7:ActionToObjectProperty Transformation Rule on rel And relStart And relTarget
        Declare relationship between individual of relStart and individual of relTarget
    else if type == "AssociateAttribute" then
        for each aa in rel do
            relStart = aa.ParentPart
            relTarget = aa.DataType
            if aa.type == "Item" then
                Apply Rule 8:CompositionAssociateAttToObjectProperty Transformation Rule on relStart And relTarget
                Declare relationship between individual of relStart and individual of relTarget
            else if aa.type == "ref Part" then
                Apply Rule 9:NotCompositionAssociateAttToObjectProperty Transformation Rule on relStart And relTarget
                Declare relationship between individual of relStart and individual of relTarget
            end if
        end for
    end if
end for
```

Fig. 9. Algorithm 3 for relationship transformation.

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Table 2. SWRL rules that associate individuals of different ontologies

No.	SWRL rules
S1	Scenario(?s),involvesElement(?s,?ise),equivalentElement(?ise,?e) -> involvesElement(?s,?e)
S2	Scenario(?s),HazardElement(?he),involvesElement(?s,?he),startTime(?he,?st) -> startTime(?s,?st)
S3	Scenario(?s),involvesElement(?s,?a),implementationCondition(?a,?ic),equal(?ic,true),endTime(?a,?ed) -> endTime(?s,?ed)
S4	Scenario(?s),involvesElement(?s,?hv),position(?hv,?p) -> position(?s,?p)
S5	Scenario(?s),HazardElement(?he),involvesElement(?s,?he),RolloverCondition(?he,?hre),equal(?hrc,true),SafetyFactors(?sf) -> emergencyType (?sf,“vehicle-self accident”)
S6	Scenario(?s),HazardElement(?he),involvesElement(?s,?he),BreakdownCondition(?he,?hrc),equal(?hrc,true),SafetyFactors(?sf) -> emergencyType(?sf,“vehicle-self accident”)
S7	Scenario(?s),Road(?r),HazardElement(?he),involvesElement(?s,?r),involvesElement(?s,?he),hasCollideAffectOn(?he,?r),CollideCondition(?he,?hcc),equal(?hcc,true),SafetyFactors(?sf) -> DamageCondition(?r,?hcc),AffectedRoad(?r),emergencyType(?sf,“vehicle to fixed object accident”)
S8	Scenario(?s),AffectedVehicle(?av),HazardElement(?he),involvesElement(?s,?av),involvesElement(?s,?he),hasCollideAffectOn(?he,?av), CollideCondition(?he,?hcc),equal(?hcc,true),SafetyFactors(?sf) -> emergencyType(?sf,“collision acident”),DamageCondition(?av,?hcc)
S9	Scenario(?s),involvesElement(?s,?e),peopleComposition(?e,?p),Casualty(?p,?c),SafetyFactors(?st) -> casualties(?sf,?c)
S10	Scenario(?s),Road(?r),involvesElement(?s,?r),laneComposition(?r,?D),Accesibility(?1,?a),FunctionFactors(?f) -> Accessibility(?r,?a), roadPassibility(?f,?a)
S11	Scenario(?s),Road(?r),involvesElement(?s,?r),DamageCondition(?r,?dc),EconomicFactors(?e) -> roadLoss(?e,?dc)
S12	Scenario(?s),Road(?r),involvesElement(?s,?r),Facility(?f),facilityComposition(?r,?0),DamageCondition(?f,?dc),EconomicFactors(?e) -> roadLoss(?e,?dc)
S13	Scenario(?s),ResponseAction(?a),ResponseResource(?r),involvesElement(?s,?a),involvesElement(?s,?r)associatedResources(?a,?r), implementationCondition(?a,?ic),equal(?ic,true),resourceType(?r,?rt),EconomicFactors(?e) -> utilizedCondition(?r,?ic),resourceType(?e,?rt)
S14	Scenario(?s),HazardElement(?he),ResponseAction(?a),involvesElement(?s,?he),involvesElement(?s,?a),startTime(?he,?st),xsd:time(?st), startTime(?a,?ast),xsd:time(?ast),subtract(?d,?ast,?st),TimeFactors(?tf) -> responseDuration(?tf,?d)
S15	Scenario(?s),ResponseAction(?a),involvesElement(?s,?a),startTime(?a,?st),xsd:time(?st),endTime(?a,?et),xsd:time(?et),subtract(?d,?et,?st), TimeFactors(?tf) -> disposalDuration(?tf,?d)
S16	Scenario(?s),HazardVehicle(?hv),involvesElement(?s,?hv),startTime(?hv,?st),xsd:time(?st),greaterThanOrEqual(?st,“00:00:00+00:00”), lessThan(?st,“06:00:00+00:00”),TimeFactors(?tf) -> emergencyPeriod(?tf,“early morning”)
S17	Scenario(?s),HazardVehicle(?hv),involvesElement(?s,?hv),startTime(?hv,?st),xsd:time(?st),greaterThanOrEqual(?st,“06:00:00+00:00”), lessThan(?st,“12:00:00+00:00”),TimeFactors(?tf) -> emergencyPeriod(?tf,“morning”)
S18	Scenario(?s),HazardVehicle(?hv),involvesElement(?s,?hv),startTime(?hv,?st),xsd:time(?st),greaterThanOrEqual(?st,“12:00:00+00:00”), lessSThan(?st,“18:00:00+00:00”),TimeFactors(?tf) -> emergencyPeriod(?tf,“afternoon”)
S19	Scenario(?s),HazardVehicle(?hv),involvesElement(?s,?hv),startTime(?hv,?st),xsd:time(?st),greaterThanOrEqual(?st,“18:00:00+00:00”), lessThan(?st,“24:00:00+00:00”),TimeFactors(?tf) -> emergencyPeriod(?tf,“evening”)
S20	Emergency(?e),beforePhase(?bp),InvolvedAbsorptionScenario(?iabs),inPhase(?e,?bp) -> involvsScenario(?e,?iabs)
S21	Emergency(?e),duringPhase(?dp),InvolvedAbsorptionScenario(?iabs),InvolvedAdaptionScenario(?iads),inPhase(?e,?dp) -> involvsScenario(?e,?iabs),involvsScenario(?e,?iads)
S22	Emergency(?e),afterPhase(?ap),InvolvedAbsorptionScenario(?iabs),InvolvedAdaptionScenario(?iads),InvolvedAbsorptionScenario(?iabs), InvolvedRecoveryScenario(?iars),inPhase(?e,?ap) -> involvsScenario(?e,?iabs),involvsScenario(?e,?iads),involvsScenario(?e,?iars)
S23	Emergency(?e),involvsScenario(?e,?is),equivalentScenario(?is,?s) -> involvsScenario(?e,?s)
S24	Emergency(?e),AbsorptionScenario(?s),involvesScenario(?e,?s),startTime(?s,?st) -> startTime(?e,?st)
S25	Emergency(?e),RecoveryScenario(?s),involvesScenario(?e,?s),endTime(?s,?et) -> endTime(?e,?et)
S26	Emergency(?e),AbsorptionScenario(?s),involvesScenario(?e,?s),position(?s,?p) -> position(?e,?p)

1. The association rules between scenario element and scenario (Rules S1–S19)

   Rules S1–S19 serve to facilitate the inferential transformation of individuals of the *ScenarioElement* ontology into individuals of the *Scenario* ontology. Rule S1 establishes associations between individuals of the *ScenarioElement* ontology and those of the *Scenario* ontology by leveraging the *involvesElement* and *equivalentElement* object properties. Rules S2–S4 map temporal and spatial information (*startTime*, *endTime*, and *position*) from the *ScenarioElement* ontology to corresponding attributes in the *Scenario* ontology. Rules S5–S19 relate the properties of the *ScenarioElement* ontology with the resilience properties of the *Scenario* ontology from four aspects: safety, function, economy, and time. In terms of safety, the properties of the *Scenario* ontology include *emergencyType* and casualties. Rules S5–S8 infer *emergencyType* property based on the behavior type and target objects of the *HazardElement* class within the *ScenarioElement* ontology. Rule S9 infers the *Casualties* property from the *Casualties* property of the *AffectedPeople* class within the *ScenarioElement* ontology. This property is true when someone is injured or dead; otherwise it defaults to false. In terms of function, the property of the *Scenario* ontology is *roadPassibility*. As shown in Rule S10, it can be based on the *Accessibility* property of the *Road* class and its associated *lane* class within the *ScenarioElement* ontology. In terms of economy, the properties of the *Scenario* ontology include *roadLoss* and *resourceType*. Rules S11 and S12 describe the association of the *DamageCondition* property of the *Road* class and *Facility* class in the *ScenarioElement* ontology with *roadLoss*. Rule S13 infers *resourceType* based on the type of response resource described in the *ResponsePlan* class within the *ScenarioElement* ontology. Lastly, in terms of time, Rules S14–S19 employ built-in SWRL functions such as *subtract*, *greaterThanOrEqual*, and *lessThan* to calculate durations and match time intervals, thereby determining the *responseDuration*, *disposalDuration*, and *emergencyPeriod* properties of the *Scenario* ontology.
2. The association rules between scenario and emergency (Rules S20–S26)

   Rules S20–S26 serve to facilitate the inferential transformation of individuals of the *Scenario* ontology into individuals of

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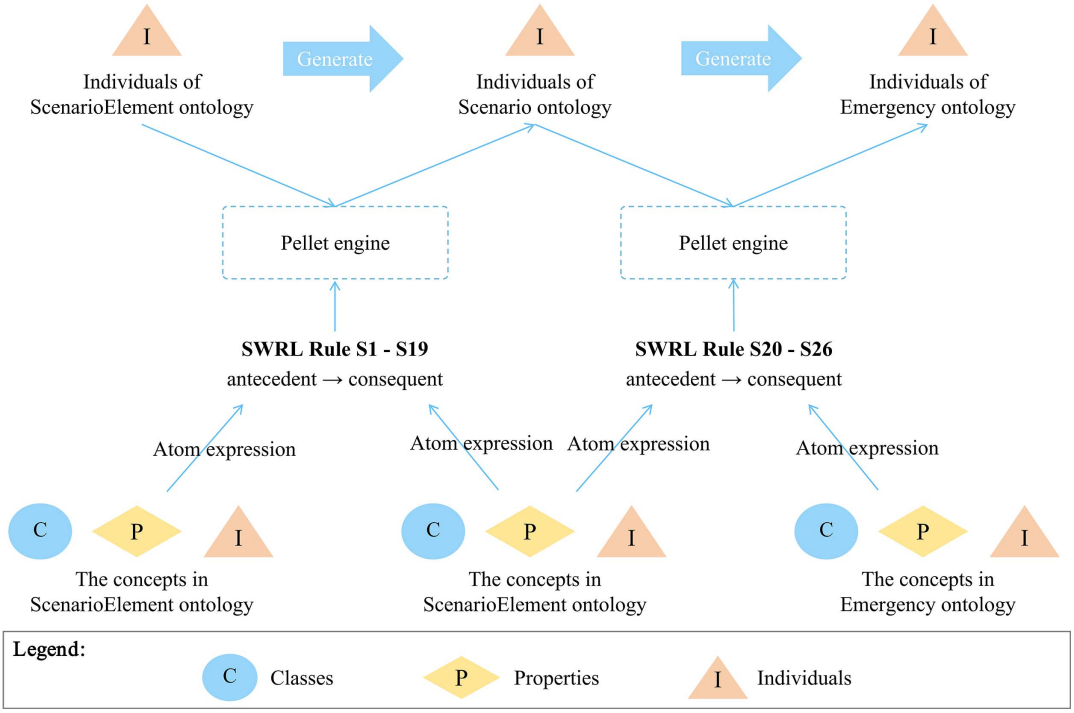


Fig. 10. Process of generating instances by executing SWRL rules through Pellet.

the *Emergency* ontology. Rules S20–S22 outline the association between resilience stages within scenarios and phases of the emergency life cycle. Specifically, when an emergency is in the *beforePhase*, *duringPhase*, or *afterPhase*, the involved scenarios are represented respectively by individuals of the *InvolvedAbsorptionScenario*, *InvolvedAdaptionScenario*, or *InvolvedRecoveryScenario* classes, which are subclasses of the *InvolvedScenario* class. Further, Rule S23 retrieves individuals of the *Scenario* ontology that are equivalent to these individuals through the *equivalentScenario* property and then associates individuals of the *Emergency* ontology with the obtained individuals. Lastly, Rules S24–S26 further facilitate the mapping of temporal and spatial information (*startTime*, *endTime*, and *position*) from the *Scenario* ontology to the corresponding properties within the *Emergency* ontology.

The above rules are executed by the Pellet engine to dynamically generate individuals, and the corresponding process is shown in Fig. 10. First, utilize Pellet to execute Rules S1–S19 with the individuals of the *ScenarioElement* ontology to generate individuals of the *Scenario* ontology. Subsequently, based on the obtained individuals, Rules S20–S26 are executed with Pellet to generate individuals of the *Emergency* ontology.

Through the above process, the ELTMs are integrated and automatically transformed into real-time ESLTM. Meanwhile, existing research has explored the integration of ontology models and BIM through IFC templates (Shi et al. 2023), which essentially involve constructing a model within a BIM environment, directly exporting it in IFC format, subsequently leveraging mature EXPRESS-to-OWL conversion tools (Pauwels and Terkaj 2016) to extract an ifcOWL ontology from the IFC file, and finally integrating this with domain ontologies. Drawing on existing methods enables the establishment of a bidirectional interaction between semantic and geometric information, thereby achieving the full-dimensional model integration of attribute, behavior, and geometry, culminating in the systematic construction of the ESCDT.

Proposed Framework for ESCDT

Based on the concept explanation and construction method for ESCDT, the overview of the proposed framework is illustrated in Fig. 11. It primarily consists of the physical layer, data layer, twin layer, knowledge layer, and service layer.

1. The physical layer aims to process information about relevant entities in the physical scenarios through data collection, mapping, and transmission. In urban road emergencies, relevant entities include the vehicle hazard element, vehicle affected element, road affected element, people affected element, road environment element, meteorology environment element, response action element, and resource element. Data acquisition equipment includes roadside equipment and external systems. Roadside equipment mainly collects various traffic parameters, vehicle types, weather, etc. External systems collect data from external channels, such as traffic and travel speed provided by the toll collection system, real-time speed and congestion conditions reported by the navigation map platform, operation plans recorded by the maintenance management system, and real-time location data provided by the rescue vehicle management system. Then, standard data interfaces are required to map these initial data into data catering to DT modeling. Moreover, real-time data transmission is made possible by edge computing and 5G technology (Liu et al. 2021). Thus, dynamic monitoring information about physical entities can be provided to the virtual space.
2. The data layer is used to collect and store twin data. The twin data can be divided into physical data, virtual data, and knowledge data (Yu et al. 2021). The physical data originate from the physical layer, which are collected by access devices. These data reflect state changes of relevant entities, furnishing a fundamental basis for emergency decision-making. The virtual data are generated through the construction and operation of twin models. On the one hand, multidimensional models are constructed using data that reflect the characteristics across various

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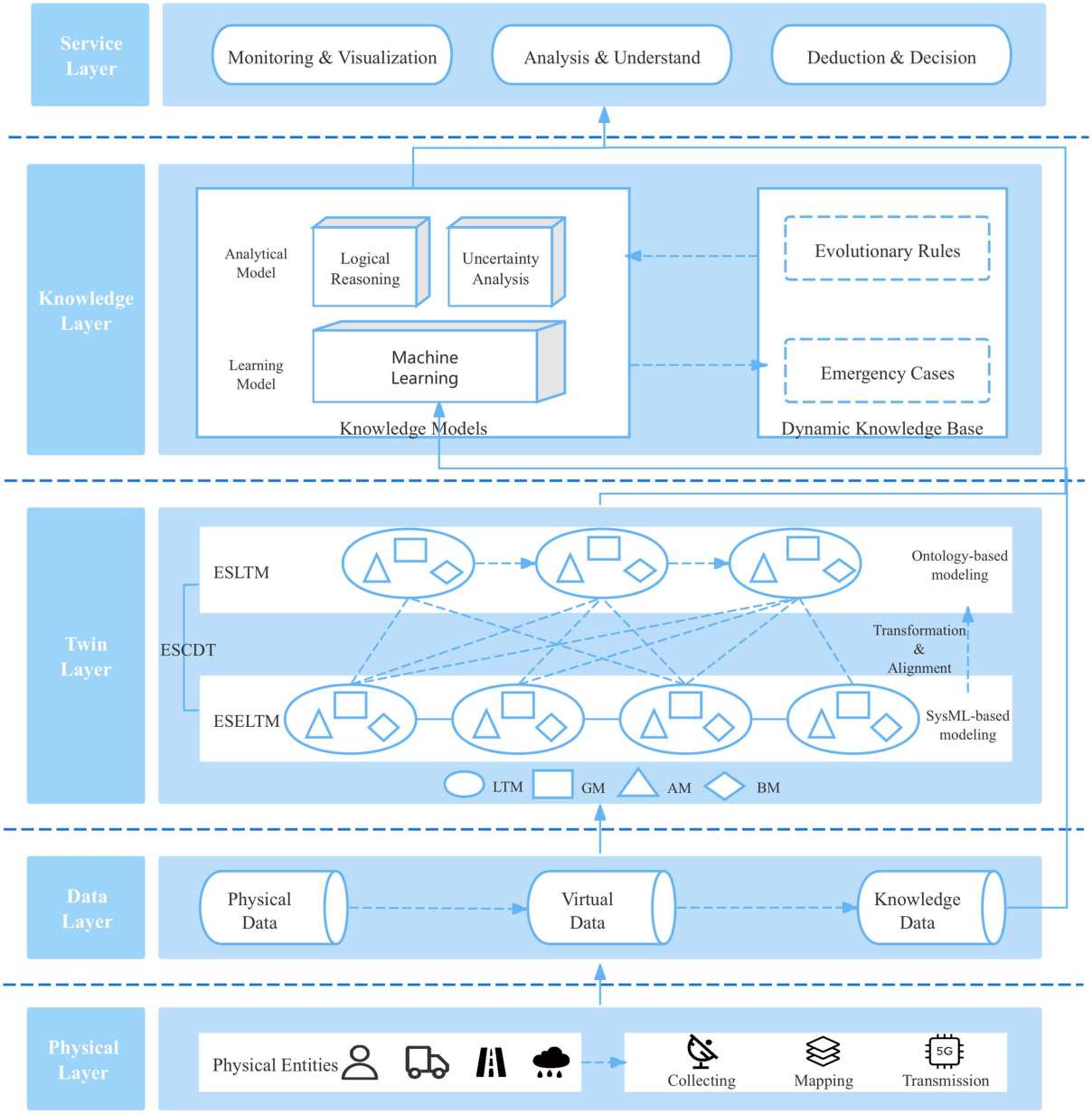


Fig. 11. Proposed framework for ESCDT.

dimensions of emergency scenario elements, which are subsequently integrated to form element-level and scenario-level twin models. On the other hand, these models generate historical data through the provision of diverse services. Furthermore, mathematical methods can be used to update current virtual data based on real-time physical data and historical twin data, such as the Bayesian method (Braik and Koliou 2024b) and the Monte Carlo method (Feng et al. 2025). The knowledge data serve the knowledge layer, which include expert knowledge and documented specifications, such as emergency plans, and emergency management laws.

3. The twin layer aims to digitally construct ESCDT with cognitive capabilities. The twin modeling is the core of this layer, as mentioned in the section "Models for ESCDT." In terms of modeling dimensions, it includes the geometry model, attribute model, and behavior model. In terms of modeling levels, it consists of ELTMs constructed by SysML and ESLTM built by ontology. These models are the virtual mappings of the physical entities based on the virtual data and could simulate and understand emergency scenarios. In addition, the ontology model enables the twin to have a rich semantic foundation and provides a shared knowledge base for cognitive services. Based on these twin models, the systematic construction of ESCDT is achieved through model transformation and geometric alignment.
4. The knowledge layer equips ESCDT with cognitive capacity by integrating a dynamic knowledge base and knowledge models that act as the brain of the cyber system. The knowledge models are composed of the learning model and the analytical model. The learning model utilizes knowledge data as the training set to update or generate knowledge through machine learning, such as the evolution rules of emergency scenarios and emergency response cases. The generated knowledge is stored in the dynamic knowledge base. The analytical model includes reasoning analysis models, and uncertainty analysis models, which utilize the knowledge in the knowledge base to interpret the current state of evolving emergency scenarios, forecast their

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development trends, and facilitate the formulation of informed emergency response decisions based on real-time scenario data.

5. The service layer encompasses multiple functional modules that provide corresponding services for different stages of accidents. Combining scenario assessment before a traffic accident occurs with immediate response afterward, which is an important step in realizing DT-based emergency management (Braik and Koliou 2024a). In this regard, based on the CDT models, both pre-event and postevent phases are benefited by providing the following services: *monitoring and visualization* service is responsible for collecting and processing real-time information on emergencies and visualizing the emergency scenarios with the help of 3D modeling technology and tools. *Analysis and understand* service conducts an in-depth analysis of monitoring data to establish a deep understanding and identification of emergency scenarios. Before a traffic accident occurs, CDT modeling helps gain in-depth insights into the structural characteristics, overall status, and evolution patterns of urban road emergency scenarios. With this knowledge, decision-makers can be warned in time and prepare for potential dangers. *Deduction and decision* service utilizes the analytical model and dynamic knowledge base to dynamically deduce the future states of emergency scenarios under the influence of different emergency plans, thereby enabling intelligent emergency decision-making. After a traffic accident, decision-makers use CDT models to simulate and evaluate the effects of various emergency plans, thereby generating new insights and refining current emergency plans to enhance the efficacy of future responses.

Case Study

Shanghai Urban Express Road Network is the backbone of transportation in Shanghai, China. The North-South Elevated Road in Shanghai Urban Express Road Network starts at the intersection of Longyao Road Tunnel and Jiyang Road and ends at the S20 Outer Ring Expressway (shown in red in Fig. 12). The total length of the road is about 22.2 km (Guan 2022), and the total flow rate of the main line is about 4,277 veh/h during the morning peak (Xu et al. 2024). It is one of the main elevated roads in Shanghai, bearing a substantial portion of the traffic and experiencing frequent traffic accidents. At about 12:25 a.m. on January 17, 2022, a box truck and a car collided on the Huma section of Gongjiang Road on the North-South Elevated Road. The accident resulted in one person being injured, garbage and oil stains appearing on the road, and damage to one anti-noise screen. After rescue, firefighting, towing, and rescue emergency units carried out rescue operations, the lane returned to normal. This study selects the above-mentioned two-vehicle collision accident (depicted in Fig. 13) as a case study to demonstrate the feasibility of the proposed ESCDT modeling method.

Construction of ELTM

This paper utilizes the SysML v2 pilot implementation to construct twin models of the scenario elements involved in the aforementioned accident. As depicted in Fig. 14, the truck is modeled as *HazardVehicle*, with *collisionCondition* and *spillCondition* attributes, as well as corresponding *collide* and *spill* behaviors. They are the key characteristics that trigger and promote the



Fig. 12. North–South elevated road. (Data from Zhang et al. 2013.)

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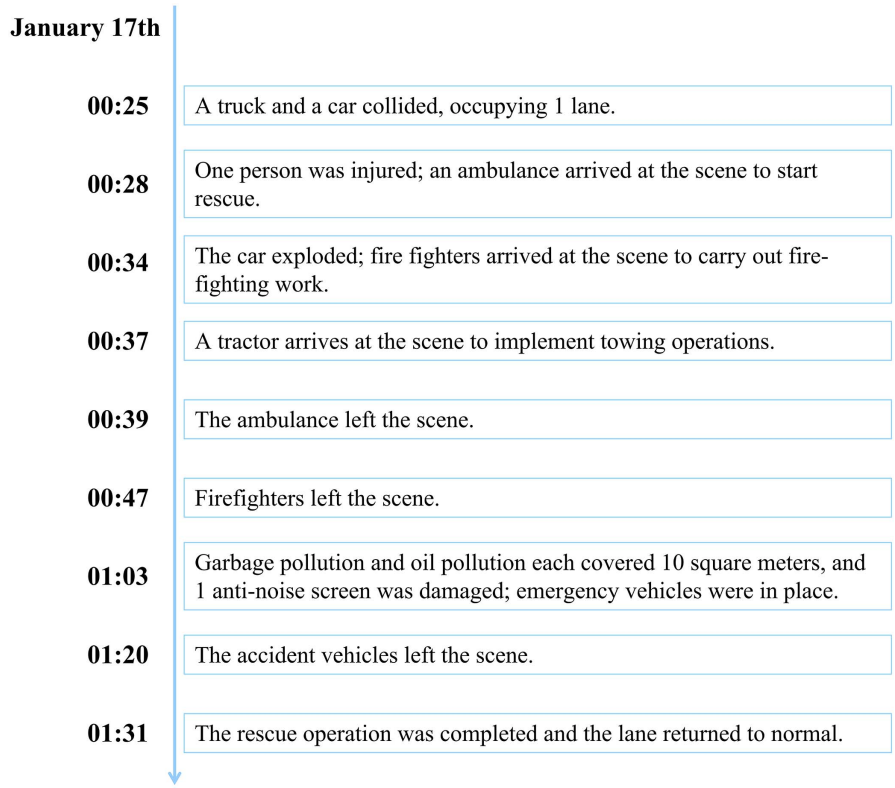


Fig. 13. Two-vehicle accident development process.

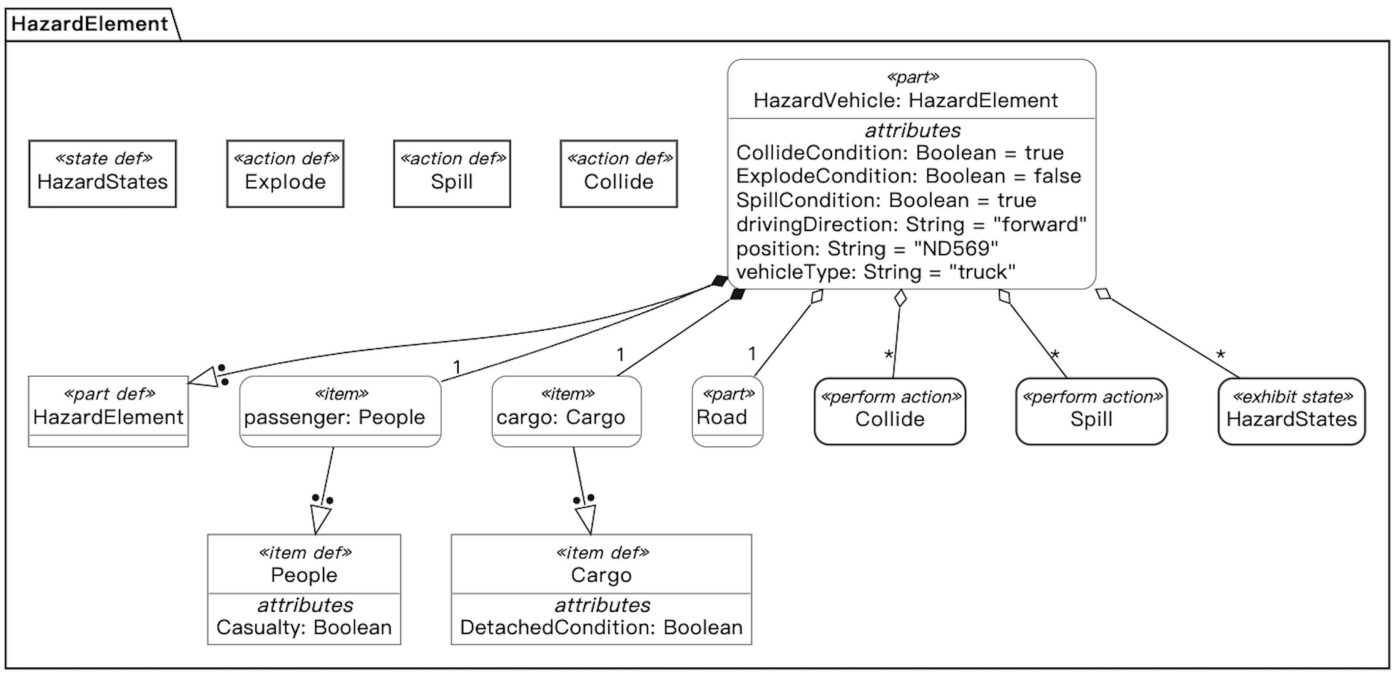


Fig. 14. Twin model of the hazard element.

occurrence and development of emergency scenarios. In Fig. 15, the car is modeled as *AffectedVehicle*, with its *damageCondition* attribute changing and subsequently emitting an *explode* behavior. Additionally, the injured passenger is modeled as *AffectedPeople*, with the *casualty* attribute representing their injury and mortality status. Furthermore, the accident results in garbage, oil pollution, and damage to noise barriers, necessitating the establishment of *AffectedRoad*, with key attributes including *damageCondition* and *pollutedCondition*. Fig. 16 illustrates the description of *Road* and *Meteorology*, respectively representing the Gongjiang Road–Huma Section and a cloudy meteorological background. With the response plan (constructed as *ResponsePlanElement*)

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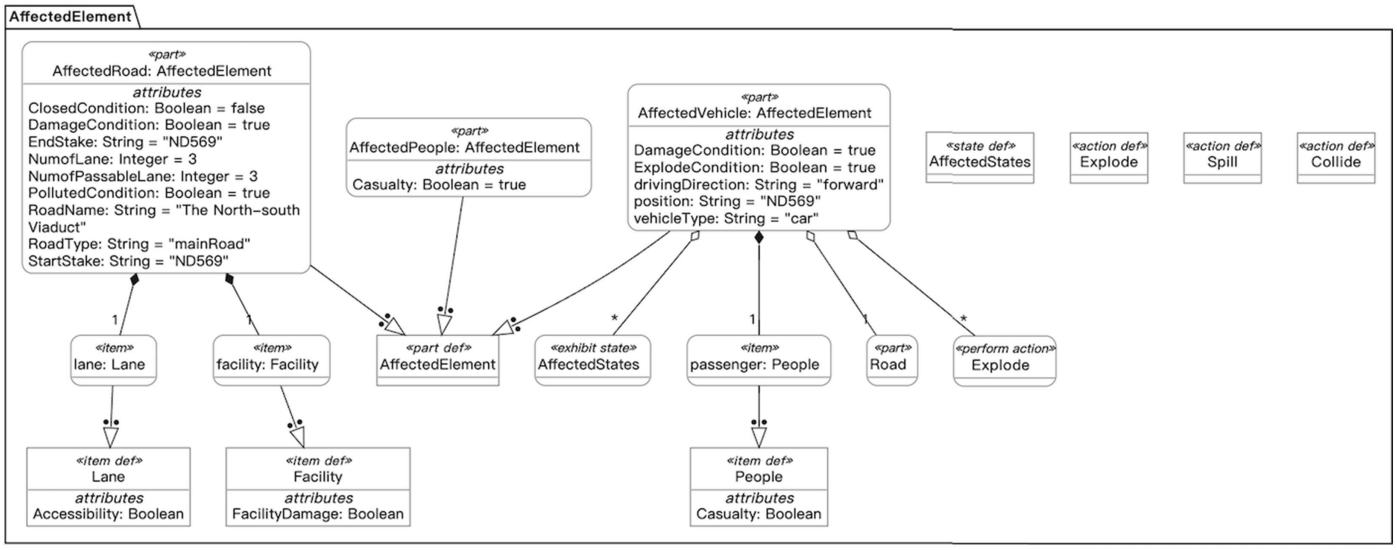


Fig. 15. Twin model of the affected element.

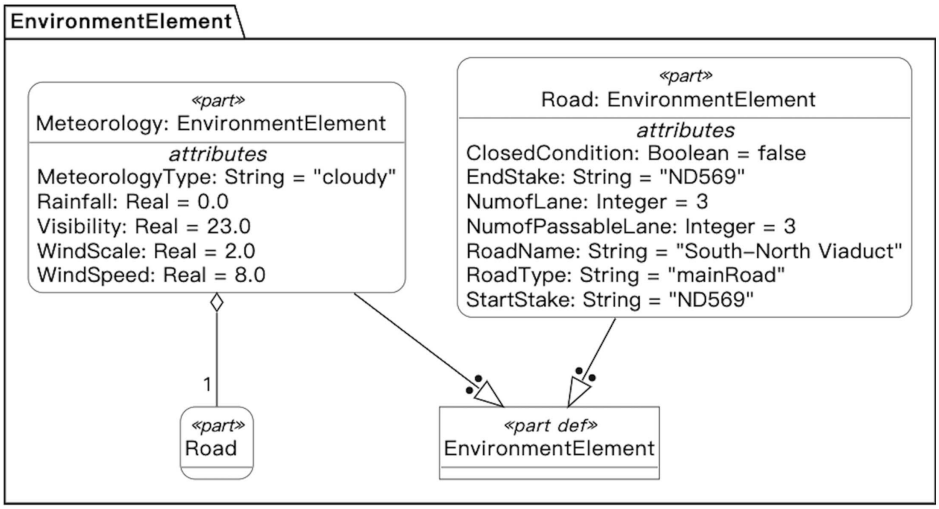


Fig. 16. Twin model of the environment element.

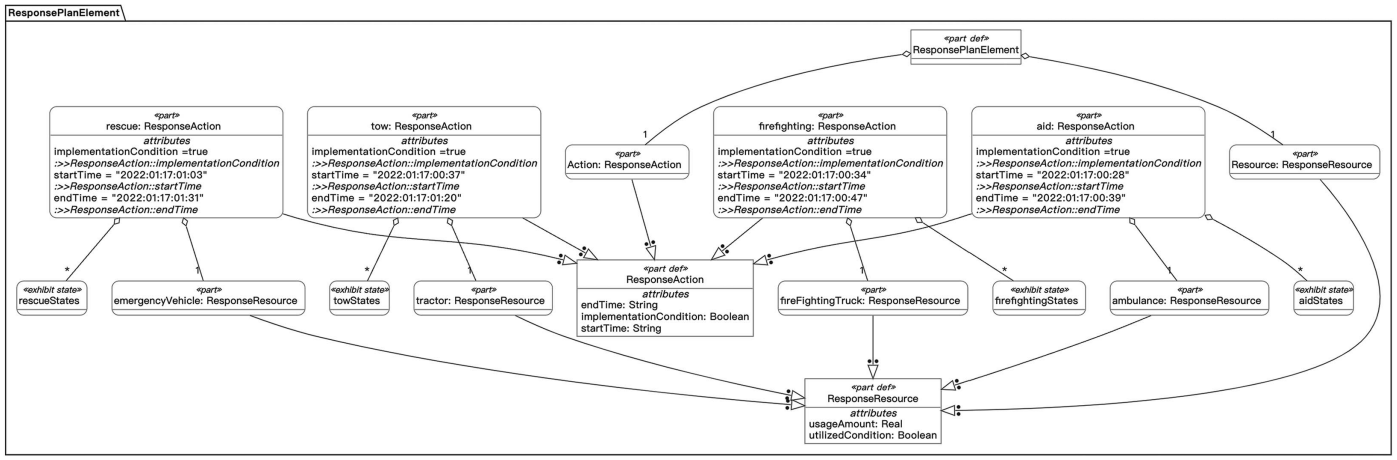


Fig. 17. Twin model of the response plan element (simplified).

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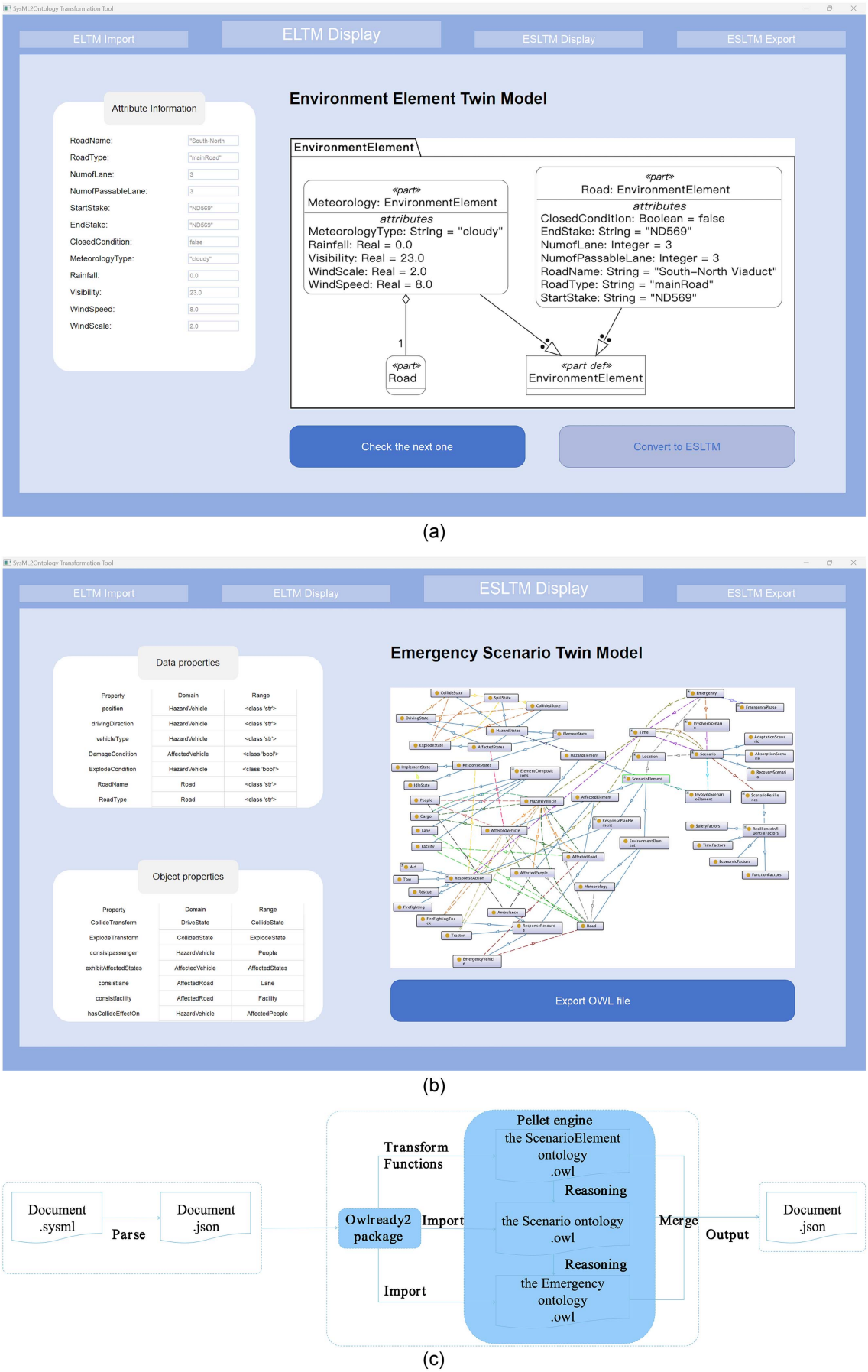


Fig. 18. (a) Interface for displaying ELTM in SysML2Ontology transformation tool; (b) interface for displaying ESLTM in SysML2Ontology transformation tool; and (c) transformation process of SysML2Ontology transformation tool.

specifications, response actions such as *aid*, *firefighting*, and *rescue* are implemented, and various response resources such as *ambulance*, *firefightingTruck*, and *emergencyVehicle* are deployed to support implementation. These actions and resources respectively constitute *ResponseAction* and *ResponseResource*, as shown in Fig. 17.

Construction of ESLTM

After constructing ELTMs, this paper developed a SysML2Ontology transformation tool based on Python 3.9 to automatically transform the ELTM constructed using SysML into an ontology model, as illustrated in Fig. 18. The core interactive interfaces are shown in Figs. 18(a and b). As shown in Fig. 18(a), ELTMs can be checked sequentially, both graphically and tabularly. After clicking the convert button, ELTMs are integrated and converted into ESLTM. ESLTM is displayed in the interface as shown in Fig. 18(b). The detailed transformation process is Fig. 18(c). This tool leverages the Owlready2 package to first take the SysML model files of ELTMs as input, parse them into JSON files, and, then, based on the model transformation rules and algorithms in the section "Modeling of ESCDT," generate the *ScenarioElement* ontology and its individuals. Additionally, it imports owl files defining the *Scenario* ontology and the *Emergency* ontology. Subsequently, the SWRL rules are executed to generate individuals of the two ontologies, ultimately integrating them to generate the owl file of ESLTM.

The ESLTM constructed utilizing the transformation tool can be opened in the Protégé software, as depicted in Fig. 19. In the *ScenarioElement* ontology, various elements, associated elements, and states are integrated respectively into subclasses of *ScenarioElement* class, *ElementCompositions* class, and *ElementState* class. In the *Scenario* ontology, the *AbsorptionScenario* class, *AdaptationScenario* class, and *RecoveryScenario* class are associated with the *ScenarioElement* ontology through *involvesElement* object property and *equivalentElement* object property. For example, the *AbsorptionScenario* class is associated with *HazardElement* class, the *AdaptationScenario* class is additionally with the *AffectedElement* class, and the *ResponseAction* class and *ResponseResource* class are associated with the *RecoveryScenario* class. The *Emergency* ontology represents the collision accident, and it is related to the *Scenario* ontology through the *involvesScenario* object property and *equivalentScenario* object property. To implement SWRL rules in Protégé, select the Pellet inference engine and then click the Start reasoner option to complete the inference and get the inference results. Part of the results are shown in Fig. 20, where potential properties (e.g. *resourceType*, *roadLoss*) are generated in the individual of *Scenario* ontology.

To verify the consistency of the generated ontology, this paper employs the HermiT reasoner. As shown in Fig. 21, the reasoner is successfully activated, and the class hierarchy remains consistent before and after reasoning, which indicates that the proposed modeling method is feasible and accurate.

Furthermore, previous work of the author's team has proposed some urban road resilience assessment methods, such as the DBN-based method (Yu et al. 2023) and the GRA-TOPSIS and LSTM-based method (Yu et al. 2024). The method proposed in this

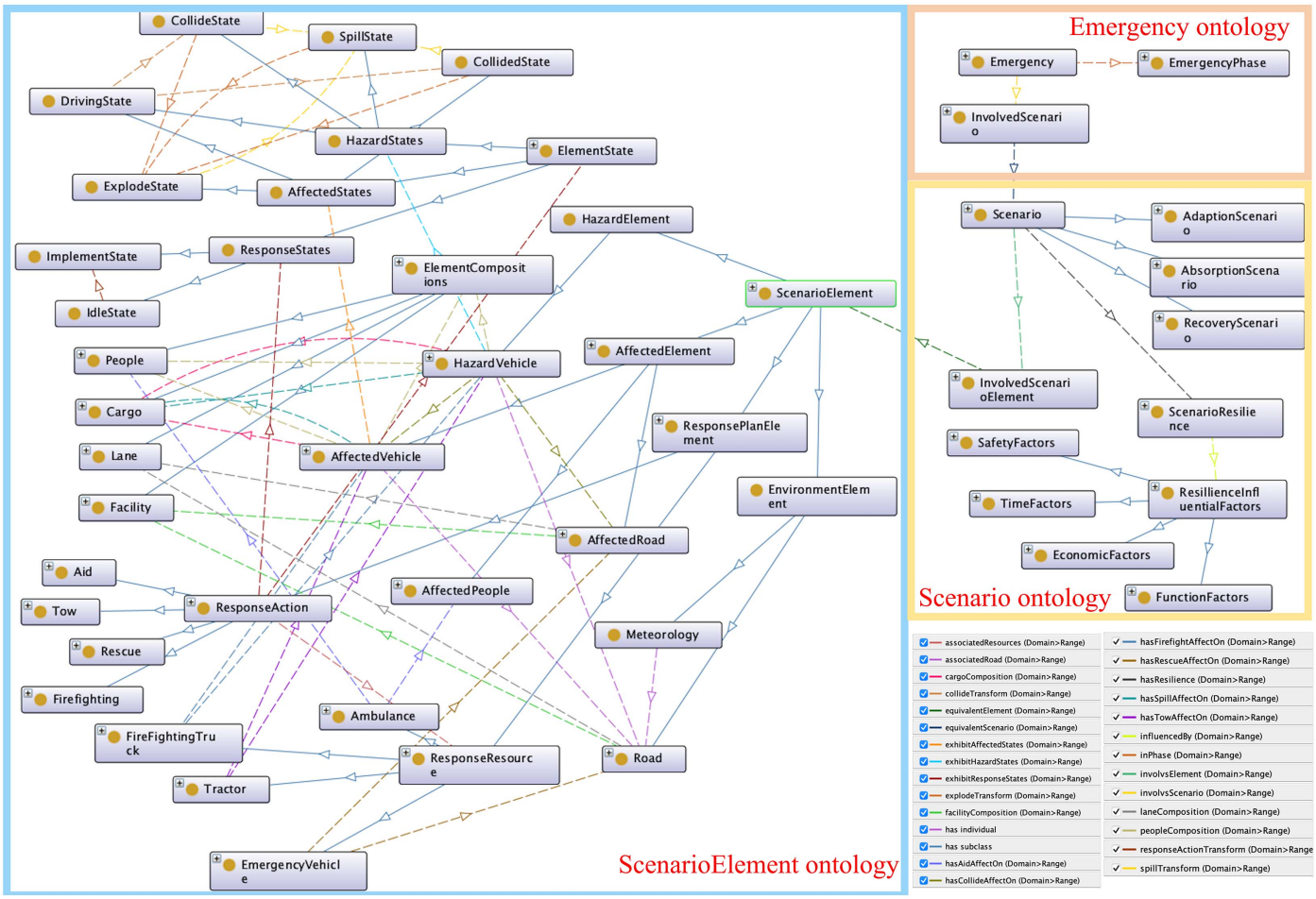


Fig. 19. ESLTM constructed with the SysML2Ontology transformation tool.

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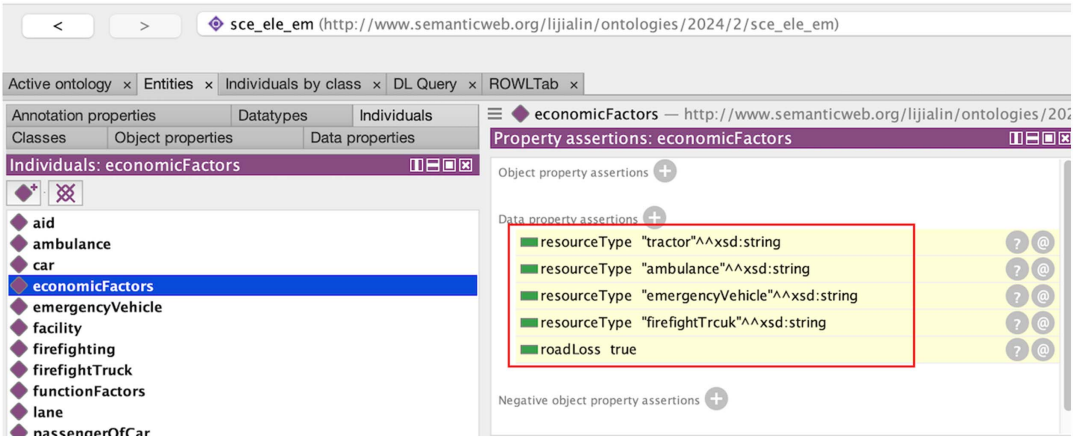


Fig. 20. Individual of *Scenario* ontology generated by reasoning SWRL rules.

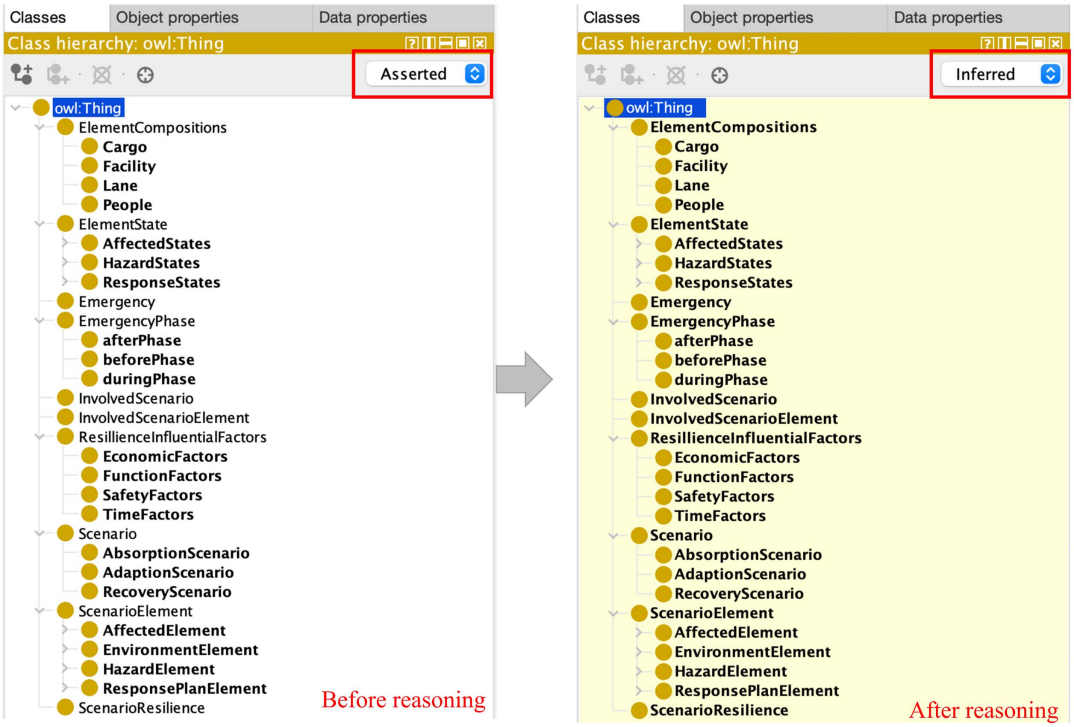


Fig. 21. Class hierarchy before and after consistency validation.

paper is capable of capturing, structurally storing, and modeling event data in real time, and, with the help of data-driven assessment methods, it can provide a dynamic and comprehensive understanding of the structural characteristics, multielement relationships, and evolving trends of urban road emergency scenarios, which is significant for promoting urban road emergency management and strengthening the resilience of the system in the pre-event and post-event phases (Zang et al. 2024).

The accuracy of DT solutions relies on the availability and utilization of large amounts of real-time data. This study validates the feasibility of the proposed approach based on historical traffic accident data set of Shanghai Urban Express Road Network (illustrated in the section "Case Study"). The SysML v2 textual notation and the standard API provide an effective means for dynamic data exchange (Bajaj et al. 2022). These features make it possible to integrate other data sources, which in turn helps narrow the gap between models and reality and reduce uncertainty. Additionally, the idea of emergency scenario analysis based on disaster theory makes it possible to construct new element-level models and form scenario-level models under larger scale or other hazard exposures by extending attribute and behavior models, which ensures the scalability of the proposed method. It is worth noting that similar analyses have been studied in larger-scale situations such as road networks (Radford et al. 2022) and communities (Nofal and van de Lindt 2022). The method proposed in this paper is currently being used in the CDT modeling of the Shanghai Urban Express Road Network.

Conclusions

Compared to DT, CDT not only achieves synchronous mapping between the physical and virtual space but also organizes and integrates data throughout the entire life cycle, enabling precise

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perception. The foundation of CDT lies in the construction of twin models with cognitive capabilities. Focusing on CDT modeling for urban road emergency management, this paper proposes a novel method for analyzing, constructing, and integrating CDT models for urban road emergency scenarios, thereby offering fresh perspectives on achieving precise mapping and comprehensive perception of emergency scenarios. The summary of the research is concluded as follows:

- This paper clarifies the definition of hierarchical twin models from the element level and scenario level, combining resilience theory to clarify the specific characteristics of different elements in emergency scenarios. By analyzing the interrelationships and couplings among multiple elements, the characteristics of twin models at different levels are described from geometry, attribute, and behavior dimensions, achieving a panoramic description of urban road emergency scenarios.
- Further, a modeling method combining SysML and ontology is proposed. The SysML-based method intuitively expresses the structure, attributes, and behaviors of scenario elements. The ontology-based method describes interactions between emergency, emergency scenarios, and scenario elements, enhancing the semantic consistency and interoperability of models. Meanwhile, this study developed the SysML2Ontology transformation tool, which enables the automatic generation of ESLTM. The feasibility of the method was verified with a collision accident.
- Additionally, this paper proposes a CDT framework tailored for urban road emergency management based on the generic CDT architecture and the requirements of urban road emergency management. This paper clarifies the definition of each layer and its key components and elucidates the relationships between the layers and components. The framework embodies the CDT operation mechanism and can guide the application in urban road emergency management.

Although this paper demonstrates the potential of the proposed approach to achieve an objective, dynamic, and comprehensive understanding of emergency scenarios in traffic accidents, it still faces challenges in directly benefiting the postevent phase. After a traffic accident, effective emergency response is critical to restoring performance and enhancing the resilience of urban road systems. Therefore, in the context of this research, future research aims to explore the scenario deduction mechanism based on the CDT models and to help discover valuable knowledge of urban road emergency response and improve the efficiency of emergency response by simulating the scenario evolution of different emergency plans under traffic accidents. These efforts will seek to provide direct support for urban road emergency response and thus enhance the digital emergency capabilities and resilience of urban transportation infrastructure.

Data Availability Statement

All data, models, or code generated or used during the study are proprietary or confidential in nature.

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